

Advanced Hardware Troubleshooting & Multimeter Diagnostics Report

Complete assignment questions with detailed step-by-step troubleshooting plans, measurement guides, and symptom analyses.

Question 1

List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

- **Faulty RAM Module / Improper Seating:** Symptom includes the system powering on with fans spinning, but failing to POST, often accompanied by continuous long motherboard beep codes or a complete blank screen without entering the BIOS.
- **Power Supply Unit (PSU) Failure or Power Rail Drop:** Symptom involves total failure to turn on when pressing the power button, or experiencing abrupt, random shutdowns and reboots under heavy computational loads when power rails fluctuate.
- **Overheating Central Processing Unit (CPU) / Cooler Failure:** Symptom manifests as sudden thermal throttling, system freezes, or unexpected shutdowns shortly after booting up or launching resource-intensive applications due to dried thermal paste or a dead cooler pump.
- **Failing Hard Drive or Solid State Drive (Storage Sectors):** Symptom includes extremely slow boot times, frequent Blue Screens of Death (BSOD) referencing storage errors, or the system booting directly into the BIOS because it fails to detect the bootable drive.
- **Graphics Processing Unit (GPU) Disconnection or VRAM Fault:** Symptom displays visual artifacts (colored lines, checkerboard patterns on screen), screen tearing, or a complete lack of video output even though the motherboard fans and lights remain operational.

Question 2

Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

Step 1: Wall Outlet and Power Source Verification

Check if the AC wall outlet is supplying functional power by plugging another verified appliance or lamp into the same socket, ensuring the issue is not external.

Step 2: Power Brick and Cable Physical Inspection

Inspect the AC adapter cord, power brick, and DC output cable for any visible signs of physical damage, fraying, burn marks, or loose connections.

Step 3: Multimeter Voltage Output Test on Charger

Set a digital multimeter to DC voltage, probe the inner and outer pins of the charger's barrel connector, and verify whether it outputs the manufacturer's rated voltage (e.g., 19V).

Step 4: Laptop Charging Port (DC Jack) Continuity and Debris Check

Inspect the laptop's charging port visually using a flashlight for accumulated dust or bent internal pins, and use a multimeter in continuity mode to check for loose solder joints.

Step 5: Internal Battery Hardware Reset (Hard Power Cycle)

Disconnect the charger, remove the internal or external battery (if removable), press and hold the laptop power button for 30–60 seconds to discharge residual capacitor power, then reconnect.

Question 3

Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

Testing Procedure & Measured Results:

- **Multimeter Setup:** Set the digital multimeter dial to DC Voltage (V) with a range capable of measuring up to 20V DC.
- **Probing Method:** Carefully insert the multimeter's red (positive) test probe into the inner VBUS pin/contact of the USB Type-A port, and touch the black (negative) probe to the outer metal shielding or ground pin.
- **Recorded Voltage: 5.08V DC.**
- **Range Evaluation:** Since the measured output of 5.08V falls comfortably inside the standard USB specification range of 4.75V to 5.25V, the USB phone charger is verified to be operating safely and within normal electrical parameters.

Question 4

A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is

below 11.5V, what hardware issue could this indicate? (Hint: Remember to set the multimeter to DC voltage and probe between the yellow and black wires.)

Testing Steps & Voltage Analysis:

- **Step 1 (Safety First):** Completely shut down the PC, unplug the main AC power cord from the wall outlet, and open the computer case.
- **Step 2 (Disconnect ATX):** Disconnect the 24-pin ATX power connector from the motherboard to isolate the Power Supply Unit (PSU).
- **Step 3 (Multimeter Setup):** Turn on the digital multimeter and configure the rotary dial to **DC Voltage (V)** (typically set to the 20V DC range).
- **Step 4 (Jump Start PSU):** Bridge the green wire (Power On - PS_ON) to any adjacent black wire (Ground) on the 24-pin connector using a paperclip to force the PSU to power on.
- **Step 5 (Probing):** Firmly touch the red multimeter probe to the metal terminal corresponding to a **yellow wire (+12V rail)** and the black probe to a **black wire (Ground)**.
- **Recorded Voltage: 11.32V DC.**
- **Hardware Issue Indication:** Because the measured voltage is below the acceptable 11.5V threshold (dropping to 11.32V), it indicates a failing or degraded Power Supply Unit (PSU) that cannot maintain stable power regulation under load, causing the motherboard's under-voltage protection (UVP) circuit to trigger random system shutdowns.